DATASCI 406 FA 26 — Module 1 Mock Exam

Name and Uniqname:

This practice exam asks about the Day 5 Benford's Law exercises. The real exam asks the same kinds of question about your project, so work this one with your worked exercises in front of you — every question points at code you have already run, though a few ask you to think past what you ran. Exercise numbers refer to the session exercise document.

No question asks you to copy a number out of your output. They ask why you did it that way, what would happen if you changed it, and where in the exercises a given technique appears.

50 points. Budget about 62 minutes.

Part A — Monte Carlo integration (10 points)

A1 (2 points)

In the exercise that draws F uniformly on $(0, 1)$, you simulated 10,000 values of X and reported the proportion carrying each leading digit. The proportion for $d = 9$ came out about 0.049, against $w_9 = 0.046$. You want a confidence interval for that 0.049. **Circle one.**

1. A binomial interval on the count of nines among the 10,000 draws.
2. A t interval across the nine proportions in the table.
3. None is needed — w_9 is exact, so the 0.049 carries no error.
4. A binomial interval on the count of nines out of 9.

A2 (2 points)

Your table of digit proportions came from 10,000 draws, and its entries are good to about two decimal places. You want a third. Roughly how many draws? **Circle one.**

1. 20,000
2. 100,000
3. 1,000,000
4. No number of draws will do it; the third decimal is beyond Monte Carlo.

A3 (6 points)

You built two samplers for $X = 10^{M+F}$. The first picks one of the nine intervals I_d uniformly and then F uniformly within it; the second takes $F \sim U(0, 1)$ directly. Both draw $M \sim \text{Poisson}(2)$.

(i) What changes in the table of digit proportions if M is drawn from Poisson(50) instead, or dropped entirely so that $X = 10^F$? Why?

(ii) The first sampler gives uniform digits and the second gives Benford digits. Name the single probabilistic difference responsible, stated in terms of the intervals I_d.

(iii) $P(D = d)$ has the closed form w_d. So what is the second simulation *for*? Say what agreement between the table and w_d establishes, and what a disagreement would mean.

Part B — variance reduction (10 points)

B1 (2 points)

To estimate $P(D = 1) = \log_{10} 2$ you draw $F \sim U(0, 1)$ and average $\mathbf{1}(F < \log_{10} 2)$. A classmate proposes antithetic pairs $(F, 1 - F)$, averaging within each pair. **Circle one.**

1. Invalid — the two draws in a pair are not independent.
2. Valid, and it cuts the variance by about 43% at a matched budget.
3. Valid, but it can never help when the quantity averaged is an indicator.
4. Valid, and it cuts the variance by about the same amount for every digit d.

B2 (2 points)

You try the antithetic pairing $(F, 1 - F)$ for every digit in turn. For one of them it makes the variance *worse*. Which, and why? **Circle one.**

1. $d = 1$, because I_1 is the widest interval.
2. $d = 3$, because I_3 is the interval containing 0.5.
3. $d = 9$, because I_9 is the narrowest interval.
4. None; the pairing can never do worse than independent draws.

B3 (6 points)

(i) Exercise 6 used $B = 4000$ and returned $p = 1/4001 \approx 0.00025$, with no null draw anywhere near the observed distance. A classmate wants to report $p = 10^{-5}$ instead. What would they have to change, and by how much? Would more FEC contributions let them report a smaller number?

(ii) The power exercise reported 0.379 at $n = 50$, from 1,000 replicates. Give the Monte Carlo standard error of that number, and say how many replicates would halve it.

(iii) That exercise spends its budget twice: one batch of replicates to set the critical value, then a fresh batch from the alternative to estimate the power. If you could afford 1,000 replicates *in total*, how would you split them? Say what goes wrong at each extreme.

Part C — Monte Carlo for estimation (10 points)

C1 (2 points)

The products exercise simulated 100,000 values of $X = \prod_{i=1}^{k} V_i$ for $k \in \{1, 5, 25, 100\}$. You want the leading digits in the table to sit closer to w_d. **Circle one.**

1. Raise the number of simulated values.
2. Raise k.
3. Either one; both shrink the gap between the table and w_d.
4. Neither; the gap is fixed by the choice of Exp(10) and cannot be changed.

C2 (2 points)

Exercise 6 simulates the null distribution with 4,000 replicates. What is *one replicate*? **Circle one.**

1. One simulated leading digit.
2. One of the nine simulated digit proportions.
3. A whole simulated dataset of 8,396 digits, reduced to a single distance.
4. The observed FEC data, resampled.

C3 (6 points)

Treat the vector of nine observed digit proportions, computed from a sample of size n, as an estimator of the true digit distribution.

(i) The $k = 1$ row puts digit 1 at about 0.95 against $w_1 = 0.301$. Is that bias in the estimator? Answer in terms of what the estimator is estimating.

(ii) In the $k = 100$ row the entries differ from w_d in the third decimal place. What would you compute to decide whether those differences are real, and roughly what value does it take for the $d = 1$ entry?

(iii) Summarize a row by the distance statistic T — the Euclidean distance between the nine observed proportions and w_d. If the truth really is Benford, is $\mathrm{E}(T) = 0$? Say why, and say what happens to the null distribution of T as n grows.

Part D — Monte Carlo hypothesis tests (10 points)

D1 (2 points)

The power exercise takes the critical value to be the 95% quantile of a batch of simulated null statistics, then draws a **fresh** batch from the alternative to estimate the power. Suppose you also wanted to check the size, and did it by asking what fraction of *those same null statistics* exceed the critical value. What do you get? **Circle one.**

1. About 0.05, confirming the test is correctly sized.
2. Exactly 0.05 up to rounding, by construction — the check establishes nothing.
3. Something well above 0.05, because reusing draws inflates the size.
4. Something well below 0.05, because a sample quantile is conservative.

D2 (2 points)

Exercise 7 takes the cutoff to be the 95% quantile of the simulated null statistics, at each n separately. What does that choice fix? **Circle one.**

1. The power, at 0.95.
2. The size, at 0.05, whatever n is.
3. The p-value for the observed data.
4. Nothing, until we see how many alternative draws exceed it.

D3 (6 points)

The test in Exercises 6–7 rejects for large T only.

(i) Why is a lower tail not part of the rejection region? Say what it would cost and what it would buy, given the alternative Exercise 7 considers.

(ii) Describe a situation in which a *small* T would be the suspicious outcome. What is the alternative hypothesis there?

(iii) A collaborator runs your test on 50 contributions, does not reject, and writes "these digits are consistent with Benford's Law, so the filings are clean." Using the Exercise 7 table, say what is wrong and what you can say instead.

Part E — the whole module (10 points)

(a) Three sample sizes (5 points)

Day 5 carries three counts that all look like a sample size. The letters standing for them have not been consistent across the module, so here they are by description:

- **the data** — 8,396 contributions in Exercise 6
- **the replicates** — 4,000 null draws in Exercise 6; in Exercise 7, 1,000 null and 1,000 alternative draws for each row of the table
- **the factors** — the $1, 5, 25, 100$ values multiplied together in Exercise 4

(i) One sentence each: what does each of the three control?

(ii) The Exercise 6 p-value is reported as $< 1/4001$. Which of the three would you raise to report a *smaller number*, and which would you raise to make the *conclusion more convincing*? These need not be the same.

(iii) Suppose the filings had held 200 contributions rather than 8,396, and the digit proportions came out about as they did. What happens to the p-value, and why? Name one number in Exercise 6 that does *not* move.

(b) Find an example (5 points)

Point to one place in the Day 5 exercises that does each of the following, and say in a sentence why it fits.

1. Estimating a probability or expectation with no closed-form answer.

2. Approximating the sampling distribution of a statistic with no tractable form.

3. Measuring an operating characteristic of a testing procedure.

4. Investigating a claim you have no formal proof of.

5. Checking a simulated answer against something you already know exactly.